Infosys Springboard Virtual Internship

“NewsPulse Global News Trend Analyzer Using AI”

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1. Introduction

In today's digital age, an enormous volume of news articles is published every day across various online platforms. With the rapid growth of information, it has become challenging for users to efficiently analyze news content, understand trends, and determine the sentiment of news articles. Automated systems that can organize, analyze, and visualize news data are therefore essential for helping users quickly extract meaningful insights.

NewsPulse is a web-based application designed to analyze and visualize news articles using **Natural Language Processing (NLP)** and **Machine Learning techniques**. The system processes news data, performs sentiment analysis, and provides interactive dashboards that help users explore news trends and sentiments. The platform classifies news articles into **Positive, Negative, and Neutral sentiments**, enabling users to understand the overall tone of current news coverage.

The application is developed using **Flask as the backend framework**, **MySQL for database management**, and various **Python libraries such as Scikit-learn, Pandas, and Matplotlib** for data processing and visualization. It incorporates features such as **user authentication, bookmarks, search history tracking, sentiment trend analysis, keyword extraction, and article export functionality**. The system also includes an **administrative dashboard** for advanced analytics, including topic modeling, TF-IDF analysis, and machine learning model training.

By combining **data analytics, machine learning, and interactive visualization**, NewsPulse provides a comprehensive platform for analyzing large collections of news articles. The project demonstrates how modern web technologies and AI-based techniques can be integrated to build intelligent applications that assist users in understanding and interpreting large-scale textual data.

2. Objective

The primary objective of the **NewsPulse** system is to develop an intelligent web-based platform capable of analyzing and visualizing news data using Natural Language Processing and Machine Learning techniques. The system aims to help users understand news sentiment, discover trending topics, and explore large collections of news articles efficiently.

The main objectives of this project are:

1. **To analyze news articles automatically** by applying Natural Language Processing techniques to extract meaningful information from textual data.
2. **To perform sentiment analysis** on news articles and classify them into three categories: Positive, Negative, and Neutral.
3. **To provide an interactive dashboard** that allows users to explore news articles, filter results, and visualize sentiment distributions.
4. **To implement secure user authentication**, allowing users to register, log in, and manage personalized features such as bookmarks and search history.
5. **To enable trend analysis and visualization**, helping users observe changes in sentiment over time and identify frequently occurring keywords.
6. **To integrate machine learning models** that can train on news data and predict sentiment for new text inputs.
7. **To allow data export and administrative analytics**, enabling deeper analysis of news datasets through features like topic modeling and TF-IDF keyword extraction.

By achieving these objectives, the system demonstrates the practical application of machine learning and web development techniques for analyzing large volumes of textual information.

3.Workflow of the System

The **NewsPulse system workflow** follows a structured pipeline that transforms raw news articles into analytical insights through multiple processing stages.

Step 1: Data Collection

The system collects real-time news articles using the **GNews API**. Multiple pages of news results are fetched and stored in a CSV dataset containing fields such as **Title, Description, Source, and Published Date**.

Step 2: Data Cleaning and Preparation

The collected news data is cleaned to remove duplicate entries, missing values, and unnecessary symbols using regular expressions. The cleaned dataset is stored as **gnews_data_cleaned.csv** for further processing.

Step 3: NLP Processing Pipeline

The cleaned news text undergoes several Natural Language Processing steps including:

- Text normalization
- Tokenization
- Stop-word removal
- TF-IDF feature extraction

These steps convert textual information into numerical vectors suitable for machine learning analysis.

Step 4: Trend Detection and Topic Modeling

To identify important themes in the news data, the system performs:

- Keyword frequency analysis
- TF-IDF ranking of important terms
- LDA topic modeling to discover hidden topics within the dataset.

Step 5: Sentiment Analysis and Model Training

The system analyzes the sentiment of news articles by combining keyword-based scoring with a **LinearSVC machine learning classifier**. The model is trained using a **train-test split and 5-fold cross-validation** to evaluate performance.

Step 6: Visualization and Analytics

The processed results are visualized using charts and dashboards that display:

- Sentiment distribution
- Trending keywords
- Topic modeling results
- Sentiment trends over time

Step 7: Web Application Interaction

The final insights are presented through a **Flask-based web application** that provides a dual portal interface:

- **User Portal:** Browse news, search articles, view sentiment analysis, and bookmark articles.
- **Admin Portal:** Perform advanced analytics such as model training, topic analysis, and exporting datasets.

This workflow enables the system to transform raw news data into structured insights, helping users better understand global news trends and sentiment patterns.

4. Code Implementation

4.1 Data Collection using GNews API

```
import requests
import csv
import re
from dotenv import load_dotenv
import os


# Load API Key

load_dotenv("gnewsapi.env")

api_key = os.getenv("API_KEY")


# Fetch News from GNews

def fetch_news(api_key):
    all_articles = []

    for page in range(1, 11):   # 10 pages × 10 articles ≈ 100
        url =
f"https://gnews.io/api/v4/search?q=technology&lang=en&max=10&page={page}&apikey={api_key}"

        response = requests.get(url)

        if response.status_code == 200:
            articles = response.json().get('articles', [])
            all_articles.extend(articles)
            print(f"Fetched page {page} → {len(articles)} articles")
        else:
            print(f"Error on page {page}: {response.status_code}")
            print(response.json())
            break

    return all_articles
```

```
# Save to CSV

def save_to_csv(articles, filename="gnews_data.csv"):
    with open(filename, mode='w', newline='', encoding='utf-8') as file:
        writer = csv.writer(file)

        writer.writerow(["Title", "Description", "Source", "Published Date"])

        for art in articles:
            writer.writerow([
                art.get('title', ''),
                art.get('description', ''),
                art.get('source', {}).get('name', ''),
                art.get('publishedAt', '')
            ])
```

4.2 Text Preprocessing

```
import pandas as pd

import re

import nltk

import numpy as np

from nltk.tokenize import word_tokenize

from nltk.corpus import stopwords

from sklearn.feature_extraction.text import TfidfVectorizer

from sklearn.decomposition import LatentDirichletAllocation

from textblob import TextBlob

# Download NLTK resources (first run only)

nltk.download('punkt')

nltk.download('stopwords')

stop_words = set(stopwords.words("english"))

# ----------------------------

# TEXT CLEANING FUNCTION

# ----------------------------

def clean_text(text):

text = str(text).lower()
```

```
text = re.sub(r'<.*?>', '', text) # Remove HTML

text = re.sub(r'[^a-zA-Z\s]', '', text) # Remove symbols

return text

# ----------------------------

# TOKENIZATION + STOPWORDS

# ----------------------------

def preprocess_text(text):

tokens = word_tokenize(text)

filtered_tokens = [

word for word in tokens

if word.isalpha()

and word not in stop_words

and len(word) > 2

]

return " ".join(filtered_tokens)

# ----------------------------

# LOAD CLEANED DATASET

# ----------------------------

df = pd.read_csv("gnews_data_cleaned.csv")

# Combine title + description (very important improvement)

df["full_text"] = df["Title"] + " " + df["Description"]

# Clean text

df["cleaned_text"] = df["full_text"].apply(clean_text)

# Tokenization + stopwords

df["processed_text"] = df["cleaned_text"].apply(preprocess_text)

print("\nText preprocessing completed.")
```

4.3 TF-IDF Feature Extraction

```
vectorizer = TfidfVectorizer(max_features=1000)

tfidf_matrix = vectorizer.fit_transform(df["processed_text"])

feature_names = vectorizer.get_feature_names_out()

print("TF-IDF Matrix Shape:", tfidf_matrix.shape)

# ----------------------------

# TRENDING KEYWORDS

# ----------------------------

mean_scores = np.mean(tfidf_matrix.toarray(), axis=0)

top_indices = mean_scores.argsort()[-10:][::-1]

print("\nTop 10 Trending Keywords:\n")

for index in top_indices:

print(feature_names[index])

# ----------------------------

# TOPIC MODELING (LDA)

# ----------------------------

lda_model = LatentDirichletAllocation(n_components=3, random_state=42)

lda_model.fit(tfidf_matrix)

print("\nDiscovered Topics:\n")

for topic_idx, topic in enumerate(lda_model.components_):

print(f"Topic {topic_idx + 1} → ", end="")

top_words = topic.argsort()[-10:][::-1]

for index in top_words:

print(feature_names[index], end=", ")

print("\n")
```

4.4 Sentiment Classification Model

```
def get_sentiment(text):

analysis = TextBlob(text)

polarity = analysis.sentiment.polarity

if polarity > 0:

return "Positive"

elif polarity < 0:

return "Negative"

else:

return "Neutral"

df["sentiment"] = df["processed_text"].apply(get_sentiment)

print("\nSentiment Distribution:\n")

print(df["sentiment"].value_counts())
```

4.5 Flask Backend API

```
from flask import Flask, jsonify, request
import pandas as pd

app = Flask(__name__)

# Load processed dataset
df = pd.read_csv("gnews_data_cleaned.csv")

# API to get overall statistics
@app.route("/api/stats")
def get_stats():
    sentiment_counts = df["sentiment_label"].value_counts()

    stats = {
        "total_articles": len(df),
        "positive": int(sentiment_counts.get("Positive", 0)),
        "negative": int(sentiment_counts.get("Negative", 0)),
        "neutral": int(sentiment_counts.get("Neutral", 0))
    }

    return jsonify(stats)


# API to fetch news articles
@app.route("/api/articles")
def get_articles():
```

```
    sentiment = request.args.get("sentiment", "All")

    filtered_df = df

    if sentiment != "All":
        filtered_df = df[df["sentiment_label"] == sentiment]

    articles = filtered_df[["Title", "Description", "Source", "Published Date"]]

    return jsonify(articles.to_dict(orient="records"))


# Run Flask server
if __name__ == "__main__":
    app.run(debug=True)
```

4.6 User Authentication

```
from flask import Flask, request, jsonify, session
import mysql.connector
import bcrypt

app = Flask(__name__)
app.secret_key = "newspulse_secret_key"

def get_db():
    return mysql.connector.connect(
        host="localhost",
        user="root",
        password="",
        database="newspulse"
    )

# User Registration
@app.route("/api/register", methods=["POST"])
def register():
    data = request.json
    username = data.get("username")
    email = data.get("email")
    password = data.get("password")

    hashed = bcrypt.hashpw(password.encode(), bcrypt.gensalt()).decode()

    conn = get_db()
    cur = conn.cursor()

    cur.execute(
        "INSERT INTO users (username, email, password) VALUES (%s,%s,%s)",
        (username, email, hashed)
    )

    conn.commit()
    cur.close()
```

```
    conn.close()

    return jsonify({"success": True, "message": "User registered successfully"})


# User Login
@app.route("/api/login", methods=["POST"])
def login():
    data = request.json
    username = data.get("username")
    password = data.get("password")

    conn = get_db()
    cur = conn.cursor(dictionary=True)

    cur.execute("SELECT * FROM users WHERE username=%s", (username,))
    user = cur.fetchone()

    cur.close()
    conn.close()

    if user and bcrypt.checkpw(password.encode(), user["password"].encode()):
        session["user_id"] = user["id"]
        session["username"] = user["username"]
        return jsonify({"success": True, "message": "Login successful"})

    return jsonify({"success": False, "error": "Invalid username or password"})
```

5. Explanation of code

5.1 Data Collection using GNews API

This section of the code is responsible for collecting real-world news articles using the **GNews API**. The API key is securely loaded from an environment file using the `dotenv` library to avoid exposing sensitive credentials in the code.

The `fetch_news()` function sends HTTP requests to the GNews API and retrieves multiple pages of news articles. A loop is used to fetch approximately **100 articles (10 pages × 10 articles per page)**. The API response is returned in JSON format, from which relevant fields such as **title, description, source, and published date** are extracted.

The `save_to_csv()` function stores the collected news articles into a structured CSV file named **gnews_data.csv**. This dataset serves as the foundation for further preprocessing and analysis in the NLP pipeline.

5.2 Text Preprocessing

This module prepares the collected news data for Natural Language Processing tasks. Several Python libraries such as **Pandas, NLTK, and regular expressions** are used to clean and normalize the text.

First, the dataset is loaded from the cleaned CSV file. The title and description fields are combined to create a more informative text field. The `clean_text()` function removes HTML tags, punctuation, and special characters while converting the text to lowercase.

The `preprocess_text()` function then performs **tokenization and stop-word removal** using the NLTK library. Tokens that are not meaningful (such as common stop words or very short words) are removed. The result is a cleaned and processed text column that is suitable for feature extraction and machine learning.

5.3 TF-IDF Feature Extraction

This section converts textual news data into numerical features using **TF-IDF (Term Frequency–Inverse Document Frequency)** vectorization. The `TfidfVectorizer` from Scikit-learn transforms the processed text into a numerical matrix that represents the importance of words in the dataset.

The code calculates the **mean TF-IDF score for each term** to identify the most significant keywords across the dataset. The top keywords represent trending topics within the news articles.

Additionally, **Latent Dirichlet Allocation (LDA)** is applied for topic modeling. The LDA algorithm identifies hidden thematic patterns within the dataset and groups words into several topics. Each topic consists of a set of high-weight keywords that represent common themes in the news data.

5.4 Sentiment Classification Model

The system uses a **keyword-based scoring approach combined with a LinearSVC machine learning classifier.**The `get_sentiment()` function analyzes each processed text entry and assigns a sentiment label based on the polarity score. Positive polarity indicates a positive sentiment, negative polarity indicates negative sentiment, and zero polarity is considered neutral.

The sentiment results are stored in a new column in the dataset, and the overall sentiment distribution is printed to understand how news articles are categorized.

5.5 Flask Backend API

The backend of the NewsPulse application is implemented using the **Flask web framework**, which provides RESTful API endpoints for the frontend dashboard.

The `/api/stats` endpoint calculates and returns summary statistics such as the total number of articles and the distribution of positive, negative, and neutral sentiments. These statistics are returned in JSON format and can be visualized on the dashboard.

The `/api/articles` endpoint allows users to retrieve news articles dynamically. Users can optionally filter articles based on sentiment categories. The filtered results are converted to JSON format and sent to the frontend, enabling interactive browsing and analysis of news articles.

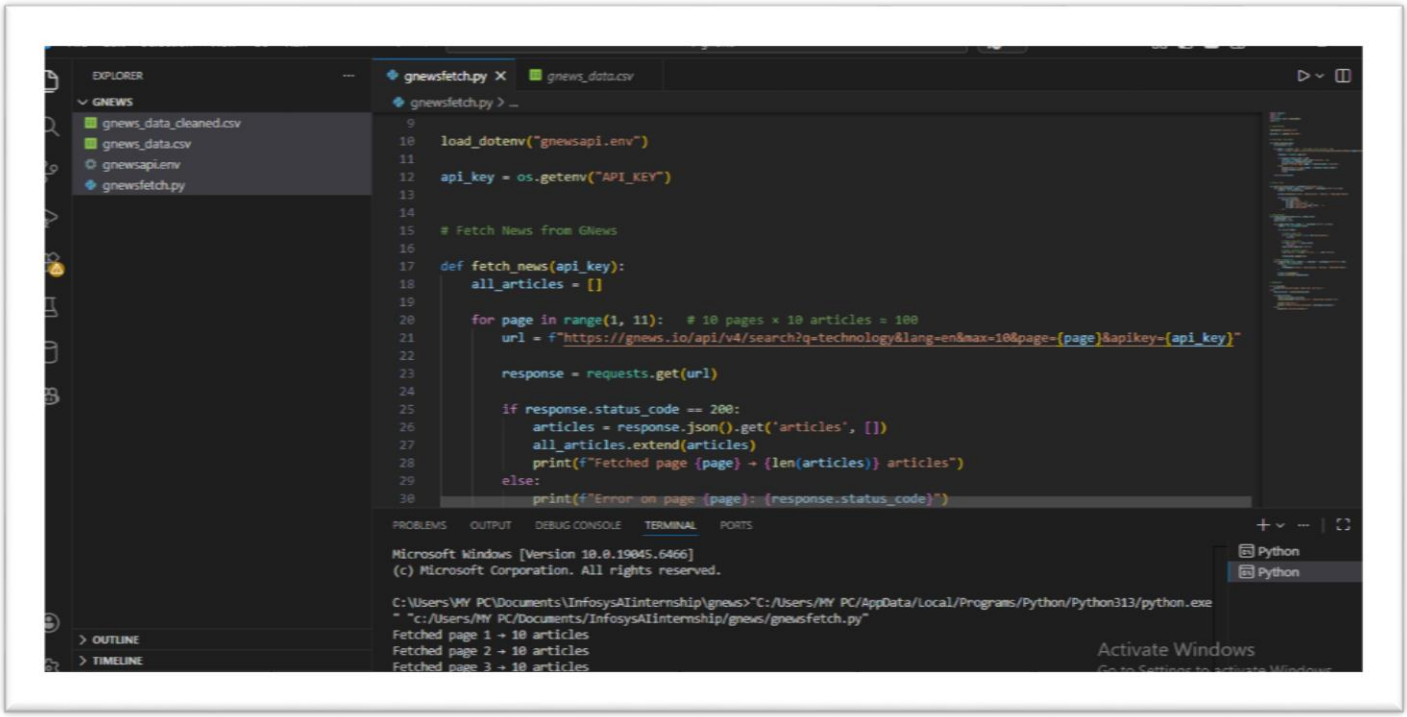
5.6 User Authentication

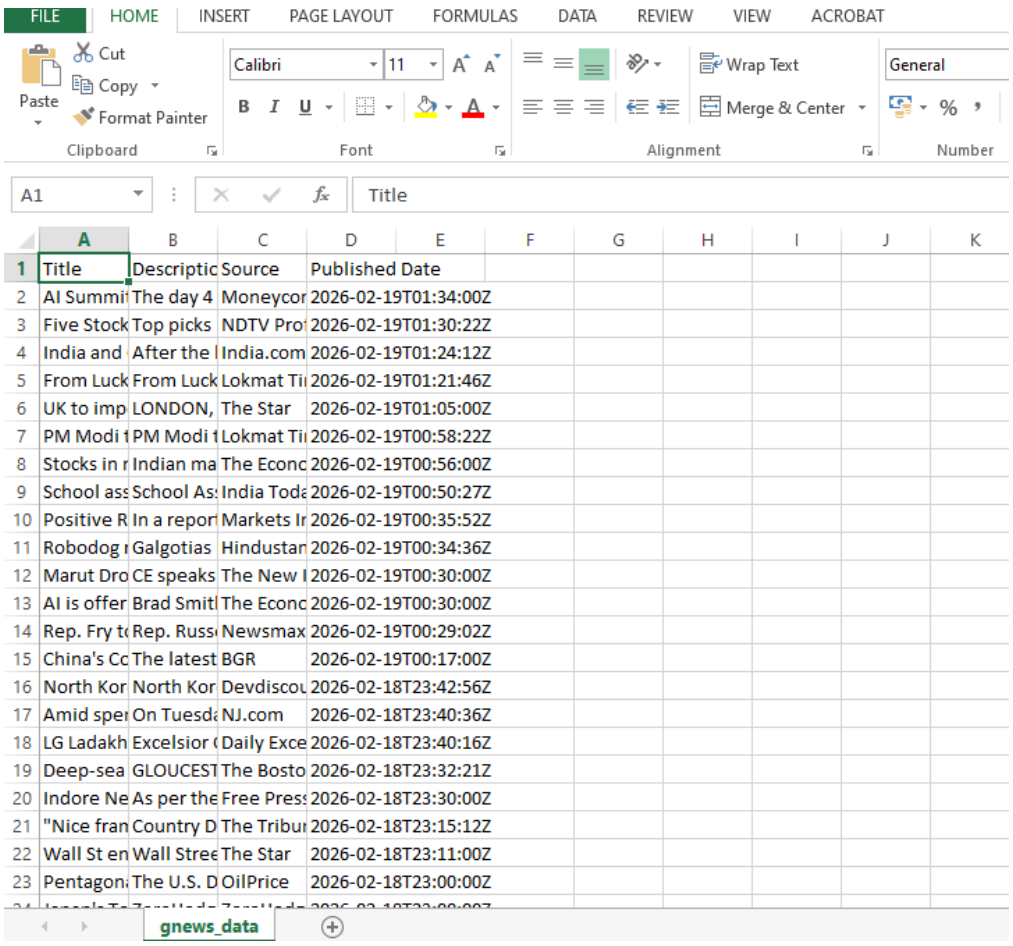
This section implements secure user authentication using **Flask, MySQL, and bcrypt password hashing**. The `get_db()` function establishes a connection to the MySQL database where user information is stored.

The registration API allows new users to create accounts by submitting their username, email, and password. The password is securely hashed using the `bcrypt` library before being stored in the database, ensuring that plaintext passwords are never saved.

The login API verifies user credentials by comparing the entered password with the stored hashed password. If authentication is successful, session variables are created to maintain the user’s login state. This allows authenticated users to access personalized features such as bookmarks and profile management within the application.

6. Output Screenshots





Title	Descriptio	Source	Published Date
AI Summit	The day 4	Moneycor	2026-02-19T01:34:00Z
Five Stock	Top picks	NDTV Pro	2026-02-19T01:30:22Z
India and	After the	India.com	2026-02-19T01:24:12Z
From Luck	From Luck	Lokmat Ti	2026-02-19T01:21:46Z
UK to imp	LONDON,	The Star	2026-02-19T01:05:00Z
PM Modi t	PM Modi t	Lokmat Ti	2026-02-19T00:58:22Z
Stocks in r	Indian ma	The Econc	2026-02-19T00:56:00Z
School ass	School As	India Toda	2026-02-19T00:50:27Z
Positive R	In a report	Markets Ir	2026-02-19T00:35:52Z
Robodog r	Galgotias	Hindustan	2026-02-19T00:34:36Z
Marut Dro	CE speaks	The New I	2026-02-19T00:30:00Z
AI is offer	Brad Smit	The Econc	2026-02-19T00:30:00Z
Rep. Fry t	Rep. Russ	Newsmax	2026-02-19T00:29:02Z
China's Cc	The latest	BGR	2026-02-19T00:17:00Z
North Kor	North Kor	Devdiscou	2026-02-18T23:42:56Z
Amid sper	On Tuesda	NJ.com	2026-02-18T23:40:36Z
LG Ladakh	Excelsior (	Daily Exce	2026-02-18T23:40:16Z
Deep-sea	GLOUCEST	The Bosto	2026-02-18T23:32:21Z
Indore Ne	As per the	Free Press	2026-02-18T23:30:00Z
"Nice fran	Country D	The Tribur	2026-02-18T23:15:12Z
Wall St en	Wall Stree	The Star	2026-02-18T23:11:00Z
Pentagon:	The U.S. D	OilPrice	2026-02-18T23:00:00Z

News Analytics Dashboard

Dataset-based NLP Analysis

Dataset Preview

	Title	Description
0	AI Summit 2026 Live Updates Bill Gates to skip summit says Gates Foundation PM Mo	The day 4 of the mega ev
1	Five Stocks To Buy Kaynes Technology Axis Bank Groww Bajaj Auto And Tata Steel	Top picks include private
2	India and China move closer as Delhi reopens government contracts to Beijing after s	After the border standoff
3	From Lucknow to NASA A space odyssey	From Lucknow to NASA:
4	UK to impose 48hour online takedown rule for nonconsensual intimate images	LONDON, Feb 18 (Reuter

Dataset Stats

Total Articles: 95

Unique Sources: 52

Run NLP Analysis

technology, meity, says, summit, new, india, said, global, million, misinformation

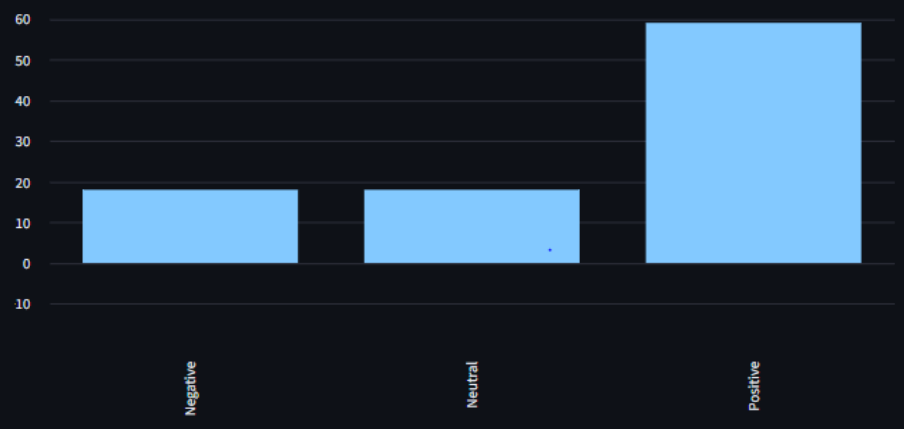
Topic 2:

technology, fishing, new, boat, try, massachusetts, authorities, company, shares, wreck

Topic 3:

technology, tech, stocks, minister, india, sector, summit, nvidia, news, feb

Sentiment Distribution



📊 News Trend Detection & Sentiment Analysis System

Dataset: gnews_data_cleaned.csv · TF-IDF + Rich Keyword Features · Target CV Accuracy ≥ 90%

✅ Dataset loaded — 95 articles | Columns: Title, Description, Source, Published Date

Preview dataset (first 5 rows)

🔥 Trend Detection

Top 10 Trending Words (Frequency)

Word	Count
the	196
and	126
to	123
technology	109
of	99

Top 10 Trending Words (TF-IDF)

Rank	Word / Phrase
1	the
2	to
3	and
4	of
5	ai

🤖 Train Sentiment Classifier

🚀 Train / Retrain Model

Class distribution:

sentiment_label	Count
Positive	39
Negative	35
Neutral	21

Cross-Val Accuracy (5-fold): 92.63% ± 4.21%

✅ Model trained! Test Accuracy: **100.00%**

📊 Model Performance

Test Accuracy	CV Mean Accuracy	CV Std Dev
100.00%	92.63%	±4.21%

Activate Windows
Go to Settings to activate Windows.

📝 Predict Sentiment for Your Own Headline

Enter a news headline or article text

India tops global innovation index with strong economic recovery

😊 Predicted Sentiment: Positive

Confidence: 70.27%

📈 Probability Breakdown

Sentiment	Probability (%)
Positive	70.27
Neutral	25.15
Negative	4.58

NewsPulse

Real-time news sentiment analysis powered by machine learning

User Portal
Browse articles, explore sentiment trends and keyword analytics

Admin Portal
Train models, run predictions and access advanced analytics

New here? Create a user account

NewsPulse
USER PORTAL

ANALYTICS
- Overview
- Articles
- Trending
- Bookmarks

ACCOUNT
- Profile

Overview

USER

TOTAL ARTICLES	POSITIVE	NEGATIVE	NEUTRAL
94	39	34	21

SENTIMENT DISTRIBUTION

TOP 10 TRENDING WORDS

SENTIMENT TREND OVER TIME

NewsPulse
ADMIN PORTAL

ANALYTICS
- Overview
- Articles
- Trending
- Bookmarks

ACCOUNT
- Profile

ADMIN TOOLS
- Word Cloud
- TF-IDF
- Topics (LDA)
- By Source
- Train Model
- Predict

Train Model

ADMIN

Train Sentiment Model

Trains a TF-IDF + LinearSVC model with keyword features and 5-fold cross-validation.

⚙ Train Model

100%	92.51%	5-fold
Test Accuracy	CV Mean ± 4.3%	Cross-Validation

Fold 1: 94.74% | Fold 2: 100% | Fold 3: 89.47% | Fold 4: 89.47% | Fold 5: 88.89%

CLASSIFICATION REPORT

CLASS	PRECISION	RECALL	F1	SUPPORT
Negative	100%	100%	100%	7
Neutral	100%	100%	100%	4
Positive	100%	100%	100%	8

CONFUSION MATRIX

	PRED NEGATIVE	PRED NEUTRAL	PRED POSITIVE
Act Negative	7	0	0
Act Neutral	0	4	0
Act Positive	0	0	8

7.Conclusion

The **NewsPulse – Global News Trend Analyzer Using AI** project successfully demonstrates how Artificial Intelligence and web technologies can be integrated to analyze and visualize large volumes of news data. The system automates the process of collecting global news articles using the GNews API, processes the textual data through a Natural Language Processing pipeline, and extracts meaningful insights such as trending topics and sentiment distribution.

The implementation includes several important components such as **data collection, text preprocessing, TF-IDF feature extraction, topic modeling, sentiment analysis, and a machine learning-based classification approach**. These techniques enable the system to analyze news content and categorize it into **positive, negative, and neutral sentiments**, helping users understand the overall tone of global news coverage.

Additionally, the project provides an **interactive web application built using Flask**, which includes both **user and admin portals**. The platform allows users to explore news articles, analyze trends, view sentiment statistics, and interact with the system through REST APIs. The integration of **secure user authentication, database storage, and dynamic data visualization** further enhances the usability of the application.

Overall, NewsPulse demonstrates the practical application of **Natural Language Processing, Machine Learning, and full-stack web development** in building an intelligent news analysis platform. The system can be further improved by integrating real-time streaming news sources, advanced deep learning models for sentiment analysis, and larger datasets to improve prediction accuracy and scalability.